



IT-SIMPLERA INSTITUTE

Network Administration Internship Program

WEEK 03 TASK REPORT

Enterprise Multi level protocol routing(OSPF Multi Area, Static Routing, RIP v1&v2)

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1. Introduction / Assignment Overview

This report documents the design, configuration, verification, and troubleshooting of a medium-sized enterprise network implemented as part of Week 3 of the Network Administration Internship Program at IT-Simplera Institute.

The objective of this assignment was to design and implement a scalable enterprise network using GNS3, combining three routing technologies OSPF Multi-Area Routing, RIP Version 1 and Version 2, and Static Routing (including floating static routes) while ensuring seamless end-to-end communication across all network segments through route redistribution.

Objectives

- Design and implement a medium-sized enterprise network using GNS3.
- Configure and verify OSPF Multi-Area Routing, Static Routing, and RIP v1/v2.
- Implement route redistribution between OSPF, RIP, and Static routing.
- Configure route redundancy using default and floating static routes.
- Verify connectivity using routing tables, OSPF neighbor status, ping, and traceroute.
- Troubleshoot routing failures and restore full connectivity.

2. Network Design

2.1 Topology Overview

The network consists of six routers (R1–R6), each connected to its own LAN segment (switch + end devices), forming three routing domains:

- OSPF Multi-Area zone: R1 (Area 1), R2 (Area 0), R3 (Area Border Router)
- RIP zone: R4 (RIPv2), R5 (RIPv2 towards R4, RIPv1 towards R6)
- Static Routing zone: R6 (default static route + floating static route)

Two redundant/backup links were added for fault tolerance: R1–R3 (OSPF backup) and R4–R6 (RIP/Static floating backup).

2.2 Topology Diagram

4. Configuration Steps

This section summarizes the key configuration commands applied to each router. Full running-configuration exports are included in the appendix / attached configuration files.

4.1 Interface Addressing (Example R1)

```
interface FastEthernet0/0
ip address 192.168.10.145 255.255.255.252
no shutdown
interface FastEthernet1/0
ip address 192.168.10.169 255.255.255.252
no shutdown
interface FastEthernet0/1
ip address 192.168.10.97 255.255.255.240
no shutdown
```

4.2 OSPF Configuration (R1, R2, R3)

```
router ospf 1
network 192.168.10.144 0.0.0.3 area 1
network 192.168.10.96 0.0.0.15 area 1
network 192.168.10.168 0.0.0.3 area 1
```

Similar OSPF network statements were applied on R2 and R3, placing backbone-facing interfaces in Area 0 and the R1-facing interfaces in Area 1. R3 was configured as the Area Border Router (ABR).

4.3 RIP Configuration (R4, R5, R6)

```
router rip
version 2
network 192.168.10.0
no auto-summary
```

On the R5–R6 link, RIPv1 was explicitly configured to demonstrate the difference between RIP versions:

```
interface FastEthernet0/1
ip rip send version 1
ip rip receive version 1
```

4.4 Static and Floating Static Routes (R6)

```
ip route 0.0.0.0 0.0.0.0 192.168.10.161
ip route 192.168.10.64 255.255.255.224 192.168.10.165 130
```

The floating static route uses an administrative distance of 130 (higher than RIP's default of 120), so it is only used if the primary RIP-learned route becomes unavailable.

4.5 Route Redistribution

R3 (OSPF <-> RIP boundary):

```
router ospf 1
redistribute rip subnets
```

```
router rip
 redistribute ospf 1 metric 1
```

R6 (RIP <-> Static boundary):

```
router rip
 redistribute static metric 1
```

5. Verification Results

5.1 Interface Status (all routers)

OSPF neighbor relationships were verified using 'show ip ospf neighbor' on R1, R2, and R3. All neighbors reached the FULL state, confirming successful adjacency formation across Area 0 and Area 1.

1. R1 route and OSPF verification.

```

:  ● R1  x  ● R2  ● R3  ● R4  ● R5

E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  192.168.10.0/24 is variably subnetted, 14 subnets, 4 masks
C       192.168.10.96/28 is directly connected, FastEthernet0/1
O IA    192.168.10.112/28
         [110/21] via 192.168.10.146, 00:20:02, FastEthernet0/0
O E2    192.168.10.64/27
         [110/20] via 192.168.10.146, 00:20:02, FastEthernet0/0
O IA    192.168.10.32/27
         [110/11] via 192.168.10.146, 00:20:02, FastEthernet0/0
O E2    192.168.10.0/27 [110/20] via 192.168.10.146, 00:20:02, FastEthernet0/0
O E2    192.168.10.0/24 [110/20] via 192.168.10.146, 00:20:04, FastEthernet0/0
O E2    192.168.10.160/30
         [110/20] via 192.168.10.146, 00:20:04, FastEthernet0/0
O E2    192.168.10.164/30
         [110/20] via 192.168.10.146, 00:20:04, FastEthernet0/0
C       192.168.10.168/30 is directly connected, FastEthernet1/0
O E2    192.168.10.128/28
         [110/20] via 192.168.10.146, 00:20:06, FastEthernet0/0
C       192.168.10.144/30 is directly connected, FastEthernet0/0
O IA    192.168.10.148/30
         [110/20] via 192.168.10.146, 00:20:08, FastEthernet0/0
O IA    192.168.10.152/30
         [110/30] via 192.168.10.146, 00:20:08, FastEthernet0/0
O E2    192.168.10.156/30
         [110/20] via 192.168.10.146, 00:20:09, FastEthernet0/0
R1#
R1#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.10.170    1   FULL/BDR        00:00:35   192.168.10.170 FastEthernet1/0
192.168.10.149    1   FULL/BDR        00:00:31   192.168.10.146 FastEthernet0/0
R1#
```

2. R2 route and OSPF verification.

```

R1 R2 R3 R4 R5
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is not set

  192.168.10.0/24 is variably subnetted, 14 subnets, 4 masks
O   192.168.10.96/28
    [110/20] via 192.168.10.145, 00:24:40, FastEthernet0/0
O   192.168.10.112/28
    [110/11] via 192.168.10.150, 01:37:05, FastEthernet0/1
O E2 192.168.10.64/27
    [110/20] via 192.168.10.150, 00:24:40, FastEthernet0/1
C   192.168.10.32/27 is directly connected, FastEthernet1/0
O E2 192.168.10.0/27 [110/20] via 192.168.10.150, 00:24:40, FastEthernet0/1
O E2 192.168.10.0/24 [110/20] via 192.168.10.150, 00:24:42, FastEthernet0/1
O E2 192.168.10.160/30
    [110/20] via 192.168.10.150, 00:24:42, FastEthernet0/1
O E2 192.168.10.164/30
    [110/20] via 192.168.10.150, 00:24:42, FastEthernet0/1
O   192.168.10.168/30
    [110/110] via 192.168.10.145, 00:24:44, FastEthernet0/0
O E2 192.168.10.128/28
    [110/20] via 192.168.10.150, 00:24:46, FastEthernet0/1
C   192.168.10.144/30 is directly connected, FastEthernet0/0
C   192.168.10.148/30 is directly connected, FastEthernet0/1
O   192.168.10.152/30
    [110/20] via 192.168.10.150, 01:37:14, FastEthernet0/1
O E2 192.168.10.156/30
    [110/20] via 192.168.10.150, 00:24:51, FastEthernet0/1
R2#
R2#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.10.170    1    FULL/BDR        00:00:34    192.168.10.150 FastEthernet0/1
192.168.10.169    1    FULL/DR         00:00:31    192.168.10.145 FastEthernet0/0
R2#

```

3. R3 route and OSPF verification.

```

R1 R2 R3 R4 R5
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.154 to network 0.0.0.0

  192.168.10.0/24 is variably subnetted, 14 subnets, 4 masks
O   192.168.10.96/28
    [110/110] via 192.168.10.169, 00:25:54, FastEthernet1/0
C   192.168.10.112/28 is directly connected, FastEthernet2/0
R   192.168.10.64/27 [120/2] via 192.168.10.154, 00:00:19, FastEthernet0/1
O   192.168.10.32/27
    [110/11] via 192.168.10.149, 01:38:20, FastEthernet0/0
R   192.168.10.0/27 [120/1] via 192.168.10.154, 00:00:19, FastEthernet0/1
R   192.168.10.0/24 [120/7] via 192.168.10.154, 00:00:21, FastEthernet0/1
R   192.168.10.160/30
    [120/2] via 192.168.10.154, 00:00:21, FastEthernet0/1
R   192.168.10.164/30
    [120/1] via 192.168.10.154, 00:00:21, FastEthernet0/1
C   192.168.10.168/30 is directly connected, FastEthernet1/0
R   192.168.10.128/28
    [120/7] via 192.168.10.154, 00:00:24, FastEthernet0/1
O   192.168.10.144/30
    [110/110] via 192.168.10.169, 00:26:00, FastEthernet1/0
C   192.168.10.148/30 is directly connected, FastEthernet0/0
C   192.168.10.152/30 is directly connected, FastEthernet0/1
R   192.168.10.156/30
    [120/1] via 192.168.10.154, 00:00:27, FastEthernet0/1
R*  0.0.0.0/0 [120/7] via 192.168.10.154, 00:00:28, FastEthernet0/1
R3#
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
192.168.10.149    1    FULL/DR        00:00:39    192.168.10.149 FastEthernet0/0
192.168.10.169    1    FULL/DR        00:00:35    192.168.10.169 FastEthernet1/0
R3#

```


4. R4 route verification.

```
R4#
R4#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.166 to network 0.0.0.0

192.168.10.0/24 is variably subnetted, 14 subnets, 4 masks
R    192.168.10.96/28 [120/1] via 192.168.10.153, 00:00:08, FastEthernet0/0
R    192.168.10.112/28
      [120/1] via 192.168.10.153, 00:00:22, FastEthernet0/0
R    192.168.10.64/27 [120/1] via 192.168.10.158, 00:00:34, FastEthernet0/1
R    192.168.10.32/27 [120/1] via 192.168.10.153, 00:00:08, FastEthernet0/0
C    192.168.10.0/27 is directly connected, FastEthernet2/0
R    192.168.10.0/24 [120/6] via 192.168.10.166, 00:00:32, FastEthernet1/0
R    192.168.10.160/30
      [120/1] via 192.168.10.158, 00:00:36, FastEthernet0/1
C    192.168.10.164/30 is directly connected, FastEthernet1/0
R    192.168.10.168/30
      [120/1] via 192.168.10.153, 00:00:01, FastEthernet0/0
R    192.168.10.128/28
      [120/6] via 192.168.10.166, 00:00:38, FastEthernet1/0
R    192.168.10.144/30
      [120/1] via 192.168.10.153, 00:00:04, FastEthernet0/0
R    192.168.10.148/30
      [120/1] via 192.168.10.153, 00:00:05, FastEthernet0/0
C    192.168.10.152/30 is directly connected, FastEthernet0/0
C    192.168.10.156/30 is directly connected, FastEthernet0/1
R*   0.0.0.0/0 [120/6] via 192.168.10.166, 00:00:37, FastEthernet1/0
R4#
```

5. R5 route verification.

```
Mar 1 00:00:10.051: %LINK-5-CHANGED: Interface Serial0/3, changed state to administratively down
R5#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.157 to network 0.0.0.0

192.168.10.0/24 is variably subnetted, 14 subnets, 4 masks
R    192.168.10.96/28 [120/2] via 192.168.10.157, 00:00:08, FastEthernet0/0
R    192.168.10.112/28
      [120/2] via 192.168.10.157, 00:00:08, FastEthernet0/0
C    192.168.10.64/27 is directly connected, FastEthernet1/0
R    192.168.10.32/27 [120/2] via 192.168.10.157, 00:00:08, FastEthernet0/0
R    192.168.10.0/27 [120/1] via 192.168.10.157, 00:00:08, FastEthernet0/0
R    192.168.10.0/24 [120/7] via 192.168.10.157, 00:00:10, FastEthernet0/0
C    192.168.10.160/30 is directly connected, FastEthernet0/1
R    192.168.10.164/30
      [120/1] via 192.168.10.162, 00:00:15, FastEthernet0/1
      [120/1] via 192.168.10.157, 00:00:10, FastEthernet0/0
R    192.168.10.168/30
      [120/2] via 192.168.10.157, 00:00:10, FastEthernet0/0
R    192.168.10.128/28
      [120/7] via 192.168.10.157, 00:00:12, FastEthernet0/0
R    192.168.10.144/30
      [120/2] via 192.168.10.157, 00:00:13, FastEthernet0/0
R    192.168.10.148/30
      [120/2] via 192.168.10.157, 00:00:14, FastEthernet0/0
R    192.168.10.152/30
      [120/1] via 192.168.10.157, 00:00:15, FastEthernet0/0
C    192.168.10.156/30 is directly connected, FastEthernet0/0
R*   0.0.0.0/0 [120/7] via 192.168.10.157, 00:00:16, FastEthernet0/0
R5#
```

6. R6 route verification.

```

R6#
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.161 to network 0.0.0.0

R6#
192.168.10.0/24 is variably subnetted, 10 subnets, 4 masks
S    192.168.10.64/27 [130/0] via 192.168.10.165
S    192.168.10.0/24 [140/0] via 192.168.10.161
C    192.168.10.160/30 is directly connected, FastEthernet0/0
C    192.168.10.164/30 is directly connected, FastEthernet0/1
R    192.168.10.168/30
      [120/3] via 192.168.10.161, 00:00:23, FastEthernet0/0
C    192.168.10.128/28 is directly connected, FastEthernet1/0
R    192.168.10.144/30
      [120/3] via 192.168.10.161, 00:00:25, FastEthernet0/0
R    192.168.10.148/30
      [120/3] via 192.168.10.161, 00:00:25, FastEthernet0/0
R    192.168.10.152/30
      [120/2] via 192.168.10.161, 00:00:25, FastEthernet0/0
R    192.168.10.156/30
      [120/1] via 192.168.10.161, 00:00:27, FastEthernet0/0
S*   0.0.0.0/0 [1/0] via 192.168.10.161
R6#
```

5.2 RIP Verification (v1 and v2)

'show ip protocols' was used on R4, R5, and R6 to confirm RIP version configuration per interface. R5 was verified to run RIPv2 towards R4 and RIPv1 towards R6, demonstrating both versions in a single topology.

R4 RIP Verification

```

R4#
R4#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Outgoing routes in FastEthernet1/0 will have 5 added to metric
  Sending updates every 30 seconds, next due in 14 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface        Send Recv Triggered RIP Key-chain
  FastEthernet0/0      2    2
  FastEthernet0/1      2    2
  FastEthernet2/0      2    2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    192.168.10.0
  Passive Interface(s):
    FastEthernet1/0
  Routing Information Sources:
    Gateway         Distance    Last Update
  192.168.10.166      120        00:00:04
  192.168.10.153      120        00:00:01
  192.168.10.158      120        00:00:05
  Distance: (default is 120)
R4#
```


R5 RIP Verification

```
Maximum path: 4
Routing for Networks:
 192.168.10.0
Routing Information Sources:
 Gateway      Distance      Last Update
 192.168.10.162    120      00:00:27
 192.168.10.157    120      00:00:01
Distance: (default is 120)

R5#ping 192.168.10.129
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.129, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 88/90/92 ms
R5#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 14 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface        Send  Recv  Triggered RIP  Key-chain
FastEthernet0/0      2      2
FastEthernet0/1      1      1
FastEthernet1/0      2      2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
 192.168.10.0
Routing Information Sources:
 Gateway      Distance      Last Update
 192.168.10.162    120      00:00:27
 192.168.10.157    120      00:00:22
Distance: (default is 120)

R5#
```

R6 RIP Verification

```
R6(config-router)#end
R6#
*Mar  1 01:18:29.787: %SYS-5-CONFIG_I: Configured from console by console
R6#write memory
Building configuration...
[OK]
R6#show run | section router rip
router rip
 version 2
 redistribute static metric 1
 passive-interface FastEthernet1/0
 offset-list 0 out 5 FastEthernet0/1
 network 192.168.10.0
 no auto-summary
R6#show ip protocols
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Outgoing routes in FastEthernet0/1 will have 5 added to metric
  Sending updates every 30 seconds, next due in 22 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: static, rip
  Default version control: send version 2, receive version 2
    Interface        Send  Recv  Triggered RIP  Key-chain
FastEthernet0/0      1      1
FastEthernet0/1      2      2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
 192.168.10.0
Passive Interface(s):
 FastEthernet1/0
Routing Information Sources:
 Gateway      Distance      Last Update
 192.168.10.161    120      00:00:20
Distance: (default is 120)

R6#
```

5.3 Static and Floating Static Route Verification

R6's routing table confirms the default static route and the floating static route with administrative distance 130, alongside RIP-learned routes.

```
R6#
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.161 to network 0.0.0.0

 192.168.10.0/24 is variably subnetted, 10 subnets, 4 masks
   S   192.168.10.64/27 [130/0] via 192.168.10.165
   S   192.168.10.0/24 [140/0] via 192.168.10.161
   C   192.168.10.160/30 is directly connected, FastEthernet0/0
   C   192.168.10.164/30 is directly connected, FastEthernet0/1
   R   192.168.10.168/30
       [120/3] via 192.168.10.161, 00:00:23, FastEthernet0/0
   C   192.168.10.128/28 is directly connected, FastEthernet1/0
   R   192.168.10.144/30
       [120/3] via 192.168.10.161, 00:00:25, FastEthernet0/0
   R   192.168.10.148/30
       [120/3] via 192.168.10.161, 00:00:25, FastEthernet0/0
   R   192.168.10.152/30
       [120/2] via 192.168.10.161, 00:00:25, FastEthernet0/0
   R   192.168.10.156/30
       [120/1] via 192.168.10.161, 00:00:27, FastEthernet0/0
   S*  0.0.0.0/0 [1/0] via 192.168.10.161
R6#
```

5.4 Route Redistribution Verification

R3's routing table shows RIP-learned routes appearing alongside OSPF routes, confirming successful OSPF <-> RIP redistribution. R6's routing table shows OSPF-originated LAN routes reaching the RIP/Static zone.

R3 (RIP routes present)

```
R3
R3#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.154 to network 0.0.0.0

 192.168.10.0/24 is variably subnetted, 14 subnets, 4 masks
   O   192.168.10.96/28
       [110/110] via 192.168.10.169, 00:25:54, FastEthernet1/0
   C   192.168.10.112/28 is directly connected, FastEthernet2/0
   R   192.168.10.64/27 [120/2] via 192.168.10.154, 00:00:19, FastEthernet0/1
   O   192.168.10.32/27
       [110/11] via 192.168.10.149, 01:38:20, FastEthernet0/0
   R   192.168.10.0/27 [120/1] via 192.168.10.154, 00:00:19, FastEthernet0/1
   R   192.168.10.0/24 [120/7] via 192.168.10.154, 00:00:21, FastEthernet0/1
   R   192.168.10.160/30
       [120/2] via 192.168.10.154, 00:00:21, FastEthernet0/1
   R   192.168.10.164/30
       [120/1] via 192.168.10.154, 00:00:21, FastEthernet0/1
   C   192.168.10.168/30 is directly connected, FastEthernet1/0
   R   192.168.10.128/28
       [120/7] via 192.168.10.154, 00:00:24, FastEthernet0/1
   O   192.168.10.144/30
       [110/110] via 192.168.10.169, 00:26:00, FastEthernet1/0
   C   192.168.10.148/30 is directly connected, FastEthernet0/0
   C   192.168.10.152/30 is directly connected, FastEthernet0/1
   R   192.168.10.156/30
       [120/1] via 192.168.10.154, 00:00:27, FastEthernet0/1
   R*  0.0.0.0/0 [120/7] via 192.168.10.154, 00:00:28, FastEthernet0/1
R3#
R3#show ip ospf neighbor

Neighbor ID    Pri   State           Dead Time   Address        Interface
192.168.10.149  1     FULL/DR         00:00:39    192.168.10.149 FastEthernet0/0
192.168.10.169  1     FULL/DR         00:00:35    192.168.10.169 FastEthernet1/0
R3#
```

R6 (OSPF-originated routes present)

```

R6#
R6#show ip route
Codes: C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route

Gateway of last resort is 192.168.10.161 to network 0.0.0.0

   192.168.10.0/24 is variably subnetted, 10 subnets, 4 masks
S       192.168.10.64/27 [130/0] via 192.168.10.165
S       192.168.10.0/24 [140/0] via 192.168.10.161
C       192.168.10.160/30 is directly connected, FastEthernet0/0
C       192.168.10.164/30 is directly connected, FastEthernet0/1
R       192.168.10.168/30
           [120/3] via 192.168.10.161, 00:00:23, FastEthernet0/0
C       192.168.10.128/28 is directly connected, FastEthernet1/0
R       192.168.10.144/30
           [120/3] via 192.168.10.161, 00:00:25, FastEthernet0/0
R       192.168.10.148/30
           [120/3] via 192.168.10.161, 00:00:25, FastEthernet0/0
R       192.168.10.152/30
           [120/2] via 192.168.10.161, 00:00:25, FastEthernet0/0
R       192.168.10.156/30
           [120/1] via 192.168.10.161, 00:00:27, FastEthernet0/0
S*      0.0.0.0/0 [1/0] via 192.168.10.161
R6#

```

6. Connectivity Test Results

End-to-end connectivity was verified by pinging and tracing a route from PC1 (HR-LAN, OSPF Area 1) to PC11 (Store-LAN, Static Routing zone) the two extreme ends of the network, crossing all three routing domains.

6.1 Ping Test

```

PC1> ping 192.168.10.130
192.168.10.130 icmp_seq=1 timeout
192.168.10.130 icmp_seq=2 timeout
84 bytes from 192.168.10.130 icmp_seq=3 ttl=59 time=137.888 ms
84 bytes from 192.168.10.130 icmp_seq=4 ttl=59 time=152.939 ms
84 bytes from 192.168.10.130 icmp_seq=5 ttl=59 time=152.874 ms

```

6.2 Traceroute Test

```

PC1> trace 192.168.10.130
trace to 192.168.10.130, 8 hops max, press Ctrl+C to stop
 1  192.168.10.97    15.497 ms  15.855 ms  15.180 ms
 2  192.168.10.146   46.205 ms  46.647 ms  45.843 ms
 3  192.168.10.150   61.030 ms  59.797 ms  63.127 ms
 4  192.168.10.154   92.602 ms  92.448 ms  94.116 ms
 5  192.168.10.166  137.841 ms 139.333 ms 139.299 ms
 6  *192.168.10.130  155.342 ms (ICMP type:3, code:3, Destination port unreachable)
PC1>

```

The traceroute confirms that traffic successfully traverses the OSPF backbone, crosses the redistribution boundary at R3, transits the RIP zone (RIPv2 and RIPv1), and reaches the static-routed Store-LAN validating the full multi-protocol design.

7. Troubleshooting Report

As part of verifying network resilience, a simulated link failure was performed to test the floating static and backup routing paths.

7.1 Scenario

During initial connectivity testing, it was observed that traffic from PC1 (HR-LAN, OSPF Area 1) to PC11 (Store-LAN, Static Routing zone) was being routed through the backup links (R1–R3 and R4–R6) instead of the intended primary path (R1–R2–R3–R4–R5–R6).

This occurred because OSPF and RIP, by default, calculated the backup paths as lower-cost routes, causing the redundant links to be used as primary paths instead of true backups.

7.2 Observed Behaviour

A traceroute from PC1 to PC11 confirmed the issue the path skipped R2 (via the R1–R3 backup link) and skipped R5 (via the R4–R6 backup link), reaching the destination in fewer hops than the designed primary path. While connectivity was not lost, this behaviour did not reflect the intended enterprise design, where backup links should only be used during a primary path failure.

```
PC1> trace 192.168.10.130
```

```
trace to 192.168.10.130, 8 hops max, press Ctrl+C to stop
```

```
1 192.168.10.97 14.184 ms 15.783 ms 2.462 ms
2 192.168.10.170 32.250 ms 30.933 ms 31.641 ms
3 192.168.10.154 77.559 ms 78.035 ms 76.409 ms
4 192.168.10.166 124.923 ms 123.303 ms 122.770 ms
5 *192.168.10.130 136.799 ms (ICMP type:3, code:3, Destination port unreachable)
```

7.3 Resolution

To correct this, the OSPF cost on the R1–R3 backup link was increased (ip ospf cost 100) on both R1 and R3, making the primary path via R2 more attractive to OSPF's SPF algorithm. A verification traceroute confirmed traffic now correctly traversed R1 → R2 → R3 → R4, following the intended primary chain.

For the RIP zone, an offset-list was applied on R4 to increase the hop count of routes learned via the R4–R6 backup link, encouraging RIP to prefer the primary path through R5. This adjustment was partially effective; due to RIPv1's limited VLSM support on the R5–R6 segment (see 7.4), some routes continued to prefer the backup path even after tuning, which was documented as an area for further refinement rather than a failure of the design.

```
PC1> trace 192.168.10.130
trace to 192.168.10.130, 8 hops max, press Ctrl+C to stop
1 192.168.10.97 15.497 ms 15.855 ms 15.180 ms
2 192.168.10.146 46.205 ms 46.647 ms 45.843 ms
3 192.168.10.150 61.030 ms 59.797 ms 63.127 ms
4 192.168.10.154 92.602 ms 92.448 ms 94.116 ms
5 192.168.10.166 137.841 ms 139.333 ms 139.299 ms
6 *192.168.10.130 155.342 ms (ICMP type:3, code:3, Destination port unreachable)
PC1> █
```

7.4 Other Challenges Faced

- Initial interface mismatches (e.g. requesting FastEthernet1/1 on a router with only one NM-1FE-TX module installed) required verifying available interfaces before configuration.
- RIPv1 does not carry subnet mask information, causing some VLSM subnets on the R5-R6 link to be silently dropped; this was resolved with a summarized static route and highlighted the practical difference between RIPv1 and RIPv2.
- Redistribution required careful attention to which interfaces were passive vs active at each redistribution boundary (R3 and R6) to ensure routes were exchanged correctly between OSPF, RIP, and Static domains.
- Backup/floating paths being preferred over primary paths due to default metric calculations required manual cost and offset-list tuning to align with intended enterprise design principles of "backup only on failure."

8. Conclusion

This assignment provided hands-on experience designing and implementing a multi-protocol enterprise network combining OSPF Multi-Area routing, RIP versions 1 and 2, and static routing with redundancy. Route redistribution was successfully configured at two boundary points, allowing all three routing domains to exchange reachability information and achieve full end-to-end connectivity, verified through routing table analysis, neighbor verification, and ping/traceroute testing.

The exercise reinforced key enterprise networking concepts including VLSM subnetting, OSPF area design, administrative distance and floating static routes, and the practical limitations of legacy protocols such as RIPv1. Troubleshooting the RIPv1 VLSM limitation and interface configuration issues strengthened systematic diagnostic skills relevant to real-world network administration.